

Soziales Schlussfolgern

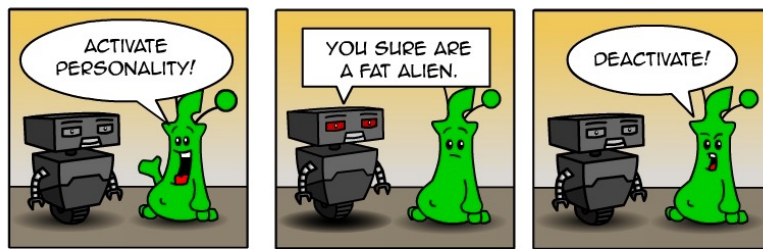


Teil 1: Zuschreiben mentaler Zustände

Teil 2: Intentionen verstehen

Teil 3: Soziales Schlussfolgern für Roboter?

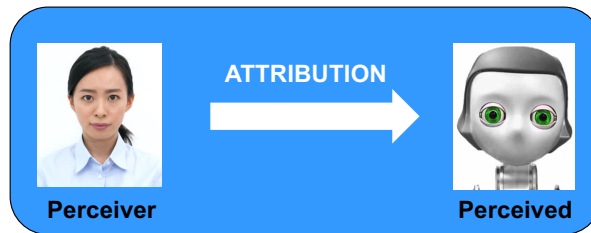
Teil 1: Zuschreiben mentaler Zustände



1. Definition von Mind Perception
2. Ursachen von Mind Perception
3. Effekte von Mind Perception

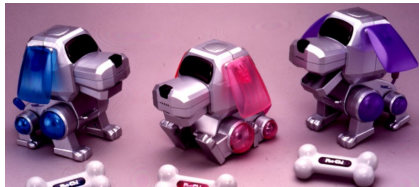
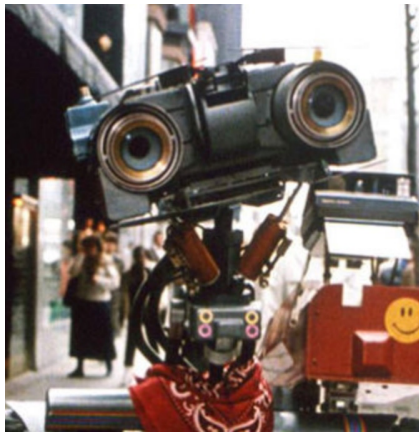
Definition von Mind Perception

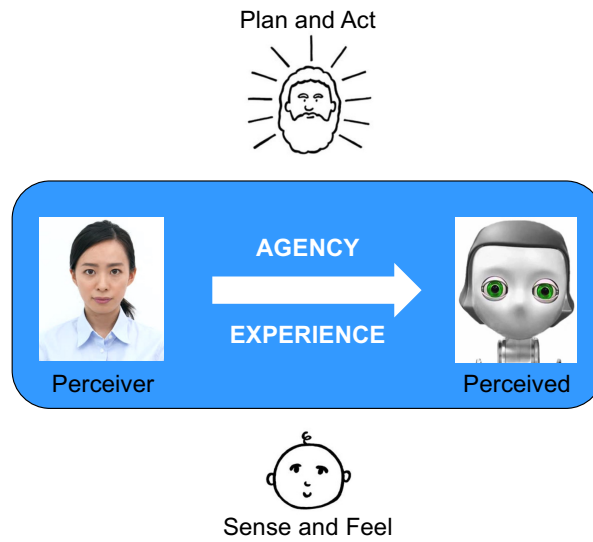


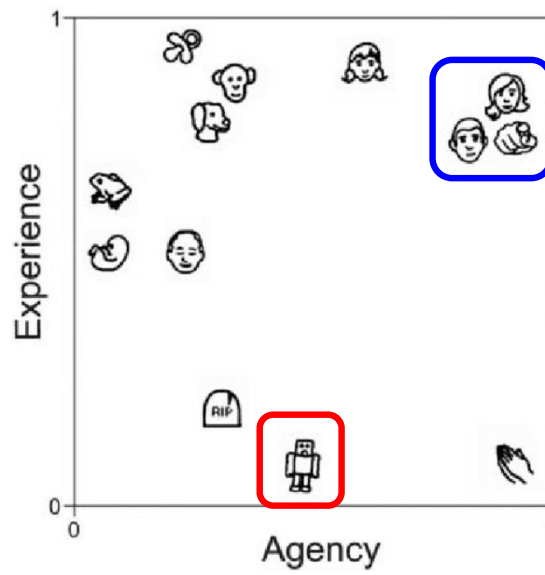


“Mind is in the Eye of the Perceiver”

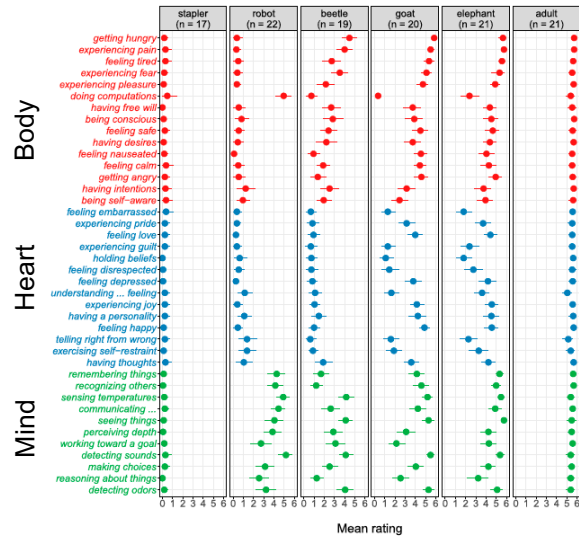
Gray et al. 2007







Gray et al. 2007



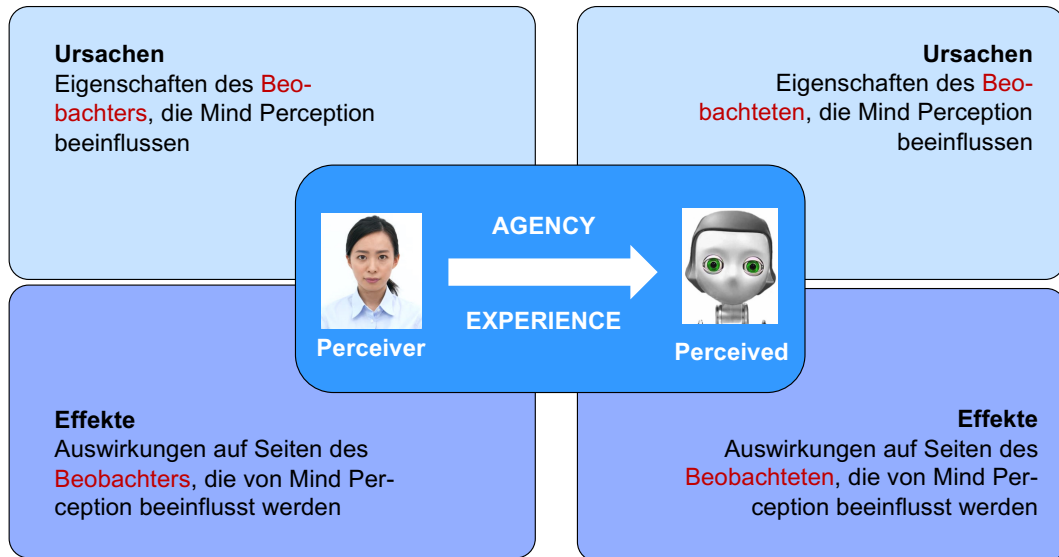
Weisman et al. 2017

<i>Desired Capacities</i>		<i>Inferred Capacities</i>	
Feeling happy	0.84	Positive Social Affect	0.96
Loving specific people	0.80		0.91
Feeling pleasure	0.79	16.2% EV	0.94
Experiencing gratitude	0.77	$\alpha = 0.88$	0.80
Feeling pain	0.86	Negative Affect	0.97
Feeling stress	0.81		0.87
Experiencing fear	0.78	15.3% EV	0.94
Feeling tired	0.77	$\alpha = 0.86$	0.95
Disapproving of immoral actions	0.80	Moral Cognition	0.83
Telling right from wrong	0.77		0.73
Upholding moral values	0.76	13.5% EV	0.84
Praising moral actions	0.66	$\alpha = 0.84$	0.81
Inferring a person's thinking	0.78	Social Cognition	0.80
Planning for the future	0.75		0.82
Understanding others' minds	0.67	12.6% EV	0.84
Setting goals	0.61	0.78	0.83
Communicating verbally	0.81	Reality Interaction	0.67
Seeing and hearing the world	0.70		0.68
Learning from instruction	0.69	12.5% EV	0.72
Moving on their own	0.68	$\alpha = 0.79$	0.76

Affect
36.7% EV
 $\alpha = 0.98$

Moral & Social Cognition
29.4% EV
 $\alpha = 0.94$

Reality Interaction
11.2% EV
 $\alpha = 0.71$



Ursachen von Mind Perception



Perceiver

- Kausale Unsicherheit
- Cognitive Load
- Wunsch nach Kontrolle
- Soziale Verbindung
- Tendenz zu Anthropomorphisieren



Perceived

Wahrgenommene Menschlichkeit

- Aussehen
- Verhalten

Similarity to Self

- Ingroup-Outgroup
- Überzeugungen

→ “**Top-down**” Merkmale
→ Wichtige **Kontextfaktoren**

→ “**Bottom-up**” Merkmale
→ Wichtige **Designfaktoren**

Waytz et al. 2010

PERCEIVER

Causal Uncertainty: Causal uncertainty is one of the basic triggers of the goal for prediction and control. When a car behaves in a perfectly predictable way in response to your actions, it seems mindless; but when it starts lurching forward while breaking, or stalling while starting, then your car might seem to have a mind of its own. This is because mental states – intentions, desires, and feelings – are the very states that best explain the behavior of independent entities

Cognitive Load: People consistently use mental states to explain both human and non-human actions, particularly when they are under cognitive load and are therefore unable to generate more elaborate causal explanations. For instance, cognitive load causes adults to explain natural events (e.g. why the sun radiates) in terms of purposeful design (e.g. to nurture life) – a teleological bias also found in young children and Alzheimer’s patients, both of whom lack well-developed causal reasoning abilities.

Lack of Control: Just as causal uncertainty increases mind perception by triggering a motivation for understanding, predictability, and control, so too does lacking personal control. For example, when people ruminate on experiences in which they lacked control or are primed with thoughts of randomness, they become more likely to believe in an agentic, controlling God, capable of planning and intention. Similarly, when reminded of existential meaninglessness and powerlessness in the face of death, people become more likely to believe in agentic Gods and other supernatural sources of agency. Supernatural agents can provide power, safety, and the potential for immortality, all of which increase one’s sense of personal meaning and control when facing death. In less existentially significant examples, simply being denied control over a set of animate marbles increases attributions of intentionality to those marbles, whereas being dispositionally inclined toward control in daily life increases the attribution of complex mental capacities to dogs.

Need for Social Connection: Other minds not only create a sense of understanding, they also create a sense of social connection with another entity. Considering another person’s mental states by explicitly adopting his or her perspective increases the perceived similarity between the self and the other person. A motivation to connect with another entity can likewise trigger thoughts about that entity’s mental states. Those with a high need to belong identify emotions from facial and vocal cues more accurately than those with a low need to belong, suggesting an increased attentiveness to mental states. Naturally-occurring or situationally- induced loneliness also causes people to ascribe mental states to pets and machines, and increases belief in the presence of invisible minds such as that of God.

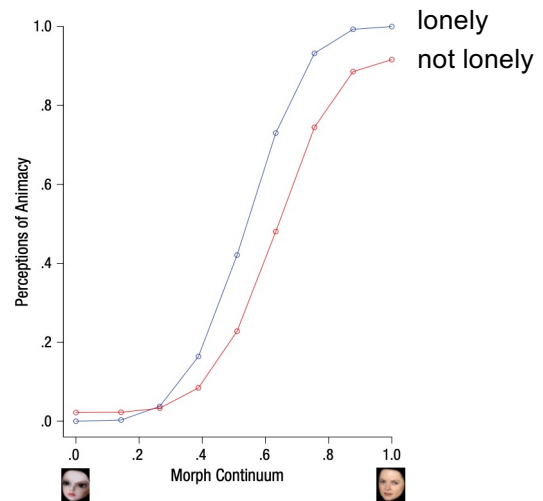
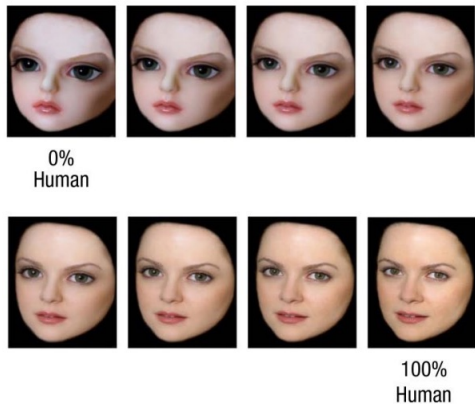
PERCEIVED

Predictability: For example, entities that act unpredictably evoke the need for control, and therefore seem more mindful than entities that behave predictably. Likewise, entities that produce negative outcomes also seem more intentional than entities that produce positive outcomes [30], such that people attribute more intentionality to others who commit evil deeds than good deeds, and express greater belief in an agentic God when attempting to understand suffering rather than salvation. This is likely because negative events elicit an increased search for causal explanations.

Human-likeness: Non-human entities that resemble humans – the prototypical mind-havers – are also attributed more mind, whether they have a humanlike appearance or simply move at a humanlike speed.

Similarity to Self: People who are similar to the self are seen as more mindful as well. In one study people judged others with similar political beliefs to be rational, whereas they judged others who held different beliefs to be less capable of ‘logical analysis’ and holding an ‘objective perspective’. Liked others also seem more mindful than disliked others. Conversely, disliked entities and those different from the self seem more mind-less. People attribute fewer secondary emotions (e.g. humiliation and nostalgia) and mental-state traits (e.g. imaginative, analytical) to out-group members than in-group members. Out-group members evoke less activation in the medial prefrontal cortex (MPFC), a region involved in perceiving other minds, and are sometimes likened to animals with diminished mental capacities. People also dehumanize (or dementalize) specific disliked others, such as another person who rejects them. Finally, when individuals objectify another person by focusing solely on the person’s body, attention is diverted from that person’s mind, making the target appear less mentally capable.

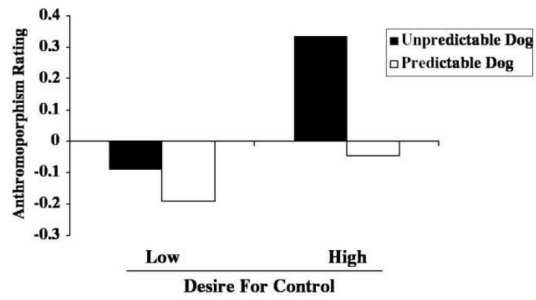




Powers et al. (2014)



© 2011 US Clin www.doggeddrawings.net

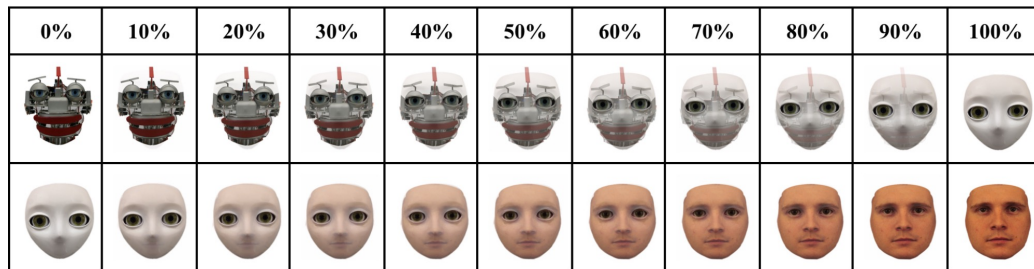


Epley et al. (2008)

Being dispositionally inclined toward control in daily life increases the attribution of complex mental capacities to dogs.



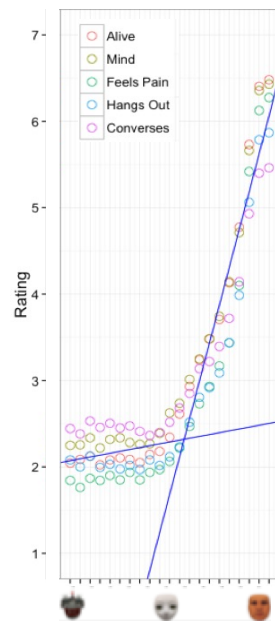
Perceived



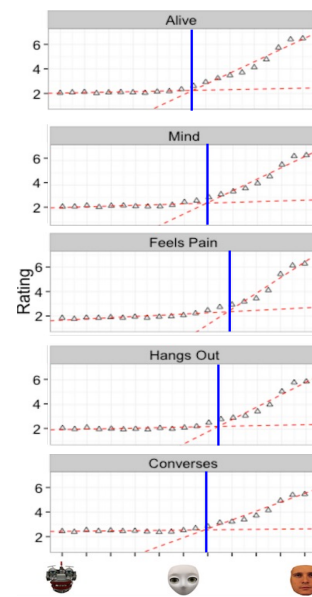
Mind Perception Fragen

- Glauben Sie, dass dieser Agent lebt?
- Hat dieser Agent einen Verstand?
- Fühlt dieser Agent Schmerz?
- Verbringt dieser Agent gern Zeit mit anderen?
- Kann man mit diesem Agenten gut Gespräche führen?

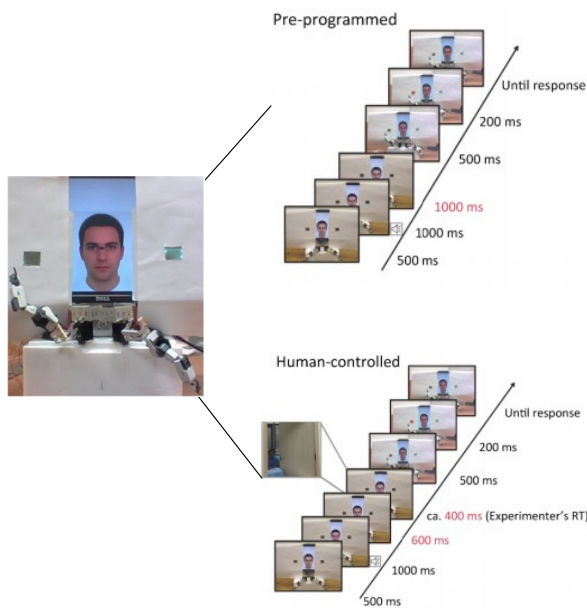
Martini, Gonzalez & Wiese (2016)



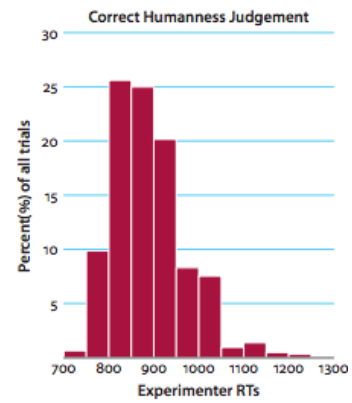
N = 160



Martini, Gonzalez & Wiese (2016)



Accuracy Level: 64%



Kajopoulos et al. (2015)

Effekte von Mind Perception



Perceiver

- Intensiviert die Erfahrung
- Verstärkt gewünschtes Verhalten
- Aktiviert soziale Stereotypen
- Moralische Rollenverteilung



Perceived

- Beeinflusst soziale Kognition
- Moralität
- Bestrafung / Schuldzuweisung
- Dehumanisierung
- Aufgabenverteilung

→ Änderung der **Einstellung**
→ **subjektive** Messmethoden

→ Änderungen im **Verhalten**
→ **objektive** Messmethoden

Waytz et al. 2010

PERCEIVER

Intensives Psychological Experience: perceiving mental states in another entity intensifies the perceiver's psychological experience of events. A seemingly random event might have little importance, but the same event intended by another could seem more significant because it comes with broader implications – intentional events are more likely to be recurrent, and demand a response. A tree branch that another person drops on you is more noteworthy than one that the wind blows down on you. People therefore experience intentional events more intensely than accidental events. An electric shock, for instance, hurts more when delivered intentionally than accidentally, and an insult intentionally directed at the self triggers more cognitive processing and rationalization than does the same insult directed at someone else. Likewise, people judge intentional harms more harshly than accidental ones.

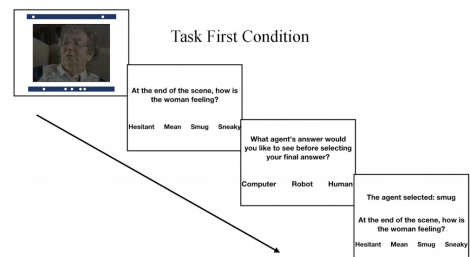
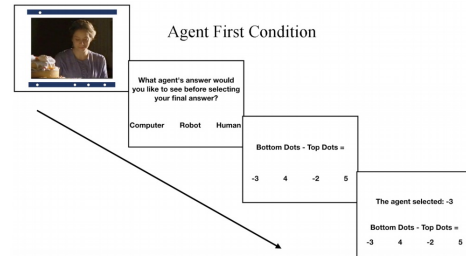
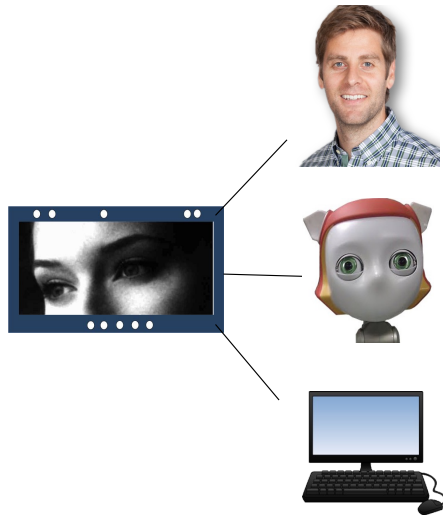
Increases Socially Desired Behavior: other minds can have thoughts, and mindful others can therefore have thoughts about the perceiver. Perceivers want to be evaluated positively by others, and the presence of another surveilling mind can therefore increase socially desirable behavior. Gods and spirits, for instance, are presumed by believers to be capable of monitoring one's own behavior at all times and in all places. Priming the presence of a mindful God therefore prompts less cheating, and more generous donations in an economic exchange. Simply evoking the external characteristics of a mind (e.g. a pair of eyes) also decreases cheating, and the presence of an audience increases people's tendency to uphold a moral code by punishing wrongdoers. Researchers have suggested that belief in such surveilling minds is adaptive because it increases prosocial behaviors that benefit one's in-group, and reduces anti-social behaviors that could lead to punishment or exile. This does not mean that mind perception is always personally beneficial. Exaggerating the presence of surveilling minds can have negative implications for the perceiver, as in the case of paranoid schizophrenics whose overactive mind perception might make them believe that they are constantly being scrutinized.

PERCEIVED

Changes Social Interactions: The perception of mind in an entity could alter how people interact with it because mind perception implies moral status. Entities capable of experience (moral patients) are afforded moral rights, whereas entities capable of agency (moral agents) are afforded moral responsibility. Given how readily people anthropomorphize, it is no surprise that people also ascribe moral responsibilities to non-human entities, such as animals and computers. In some cases, the over-attribution of moral responsibility can have harmful consequences, such as when parents ascribe intention to their infants for wrongdoing and abuse them in retribution.

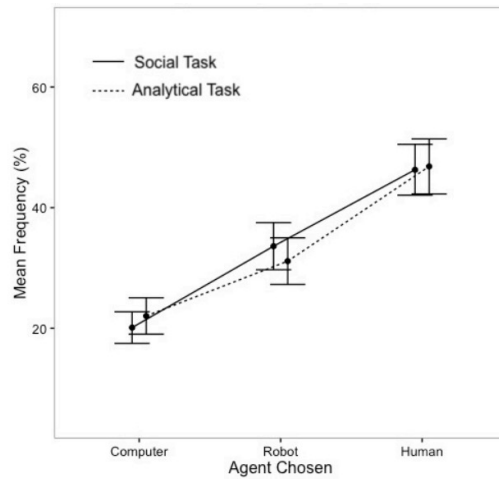
Dehumanization: The more serious ethical problem in everyday life, however, is likely to come from seeing too little mind in others. If another person is seen as relatively mindless, then he or she receives diminished moral standing, and might be treated like an animal or an object. People can deny mental capacities in two ways that map on to the dimensions of agency and experience – either people are stripped of uniquely human aspects such as competence and civility, or human nature aspects such as warmth and vitality. Each of these methods of dehumanization or demoralization has a consequence for its targets. Those denied warmth and experience come to seem robotic, cold, and cruel, encouraging active harm toward them when opportunities arise. Those denied competence, civility and agency come to be seen as subservient, or animalistic, licensing people to contain them against their will and to rob them of human rights.





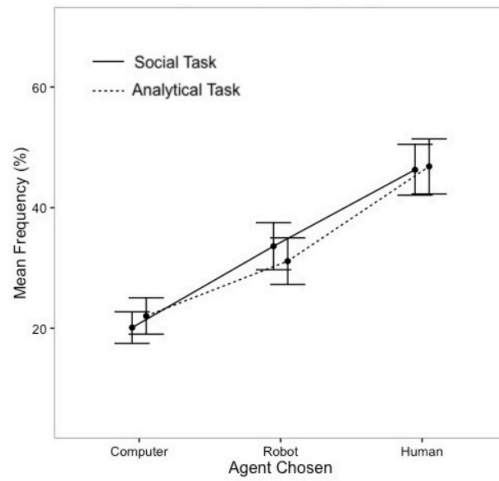
Hertz & Wiese (2018)

Agent First

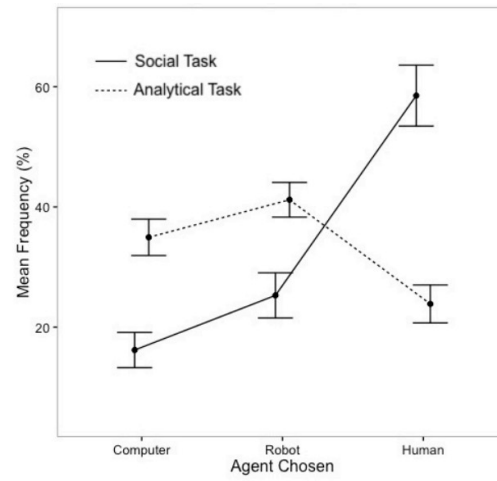


Hertz & Wiese (2018)

Agent First



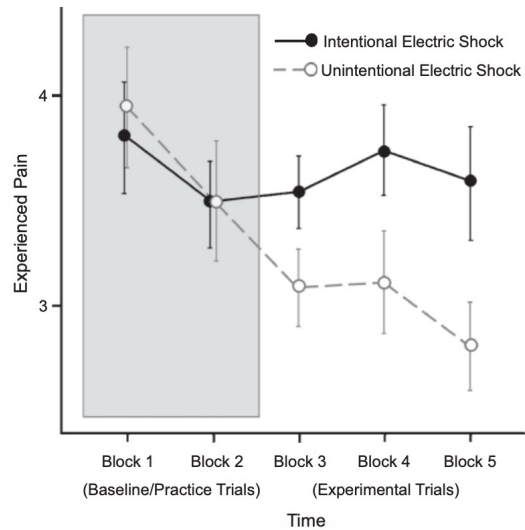
Task First



Hertz & Wiese (2018)



Manipulation:
Elektroschocks
intentional vs. aus Versehen

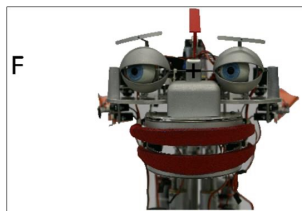
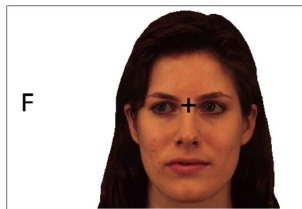


Gray & Wegner (2008)

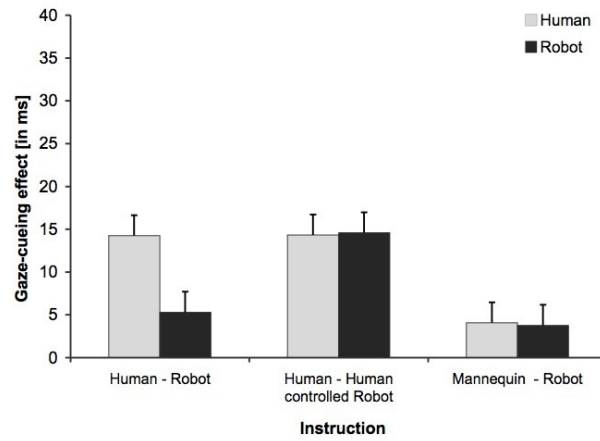


Perceived

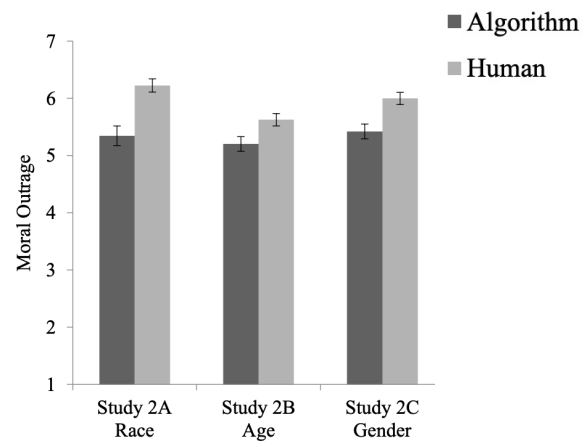
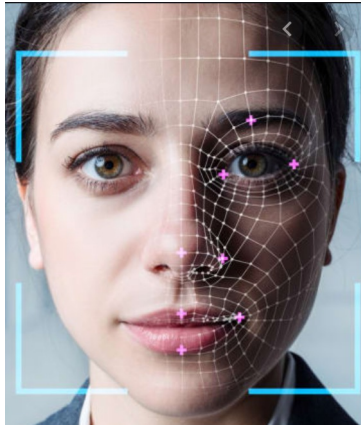
Gaze Following



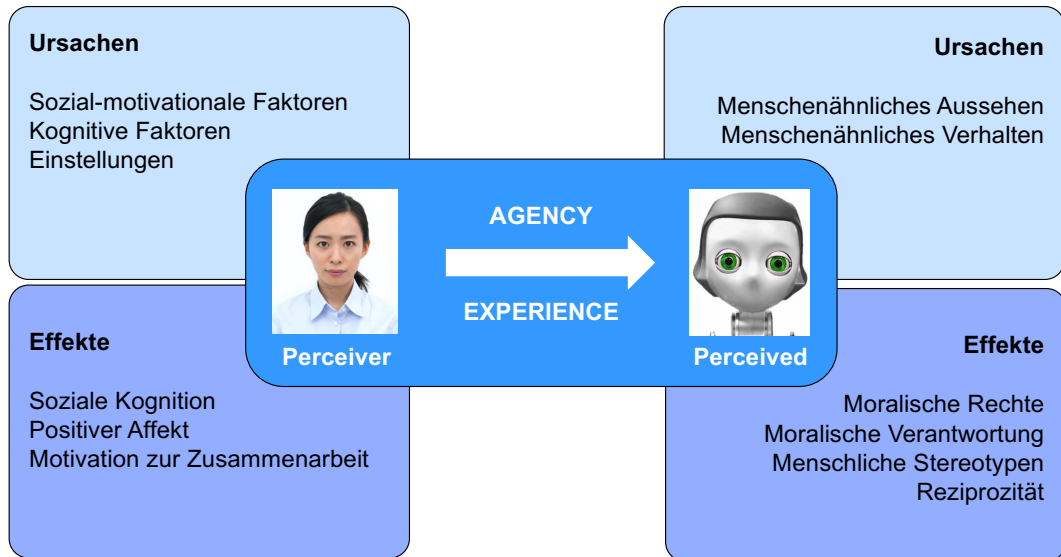
500 Durchgänge pro
Instruktion



Wiese et al. 2012

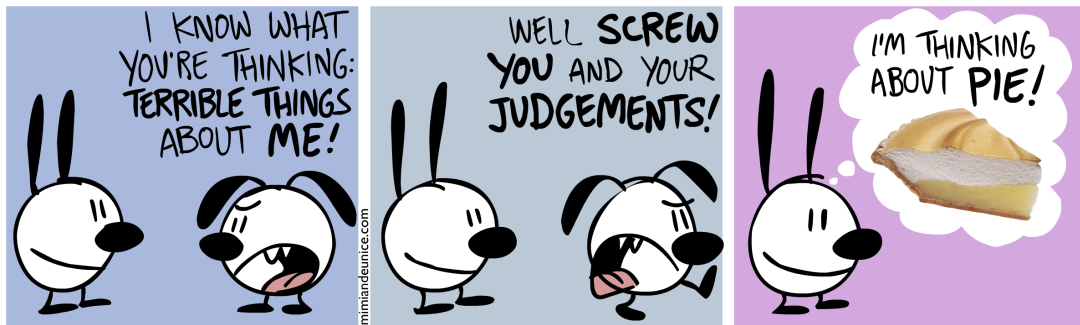


Bigman et al. (2023)



Waytz et al. 2010

Teil 2: Intentionen verstehen





Was ist soziales Schlussfolgern?

Theory of Mind (ToM)

Theory of Mind ist die **Fähigkeit**, sich selbst und anderen mentale Zustände - Überzeugungen, Intentionen, Wünsche, Behauptungen, Wissen - **zuzuordnen** und **zu verstehen**, dass Andere Überzeugungen, Wünsche und Absichten haben, die sich von den Eigenen unterscheiden.

→ ToM Defizite treten bei Menschen mit Autismus-Spektrums-Störungen, Schizophrenie, Aufmerksamkeitsdefizit-Hyperaktivitätsstörung sowie bei Neurotoxizität aufgrund von Alkoholmissbrauch auf.

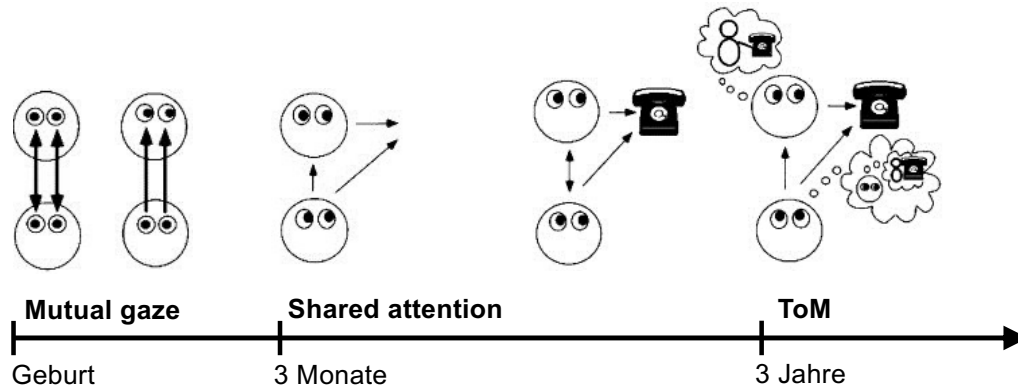
“Mind” ist nicht beobachtbar



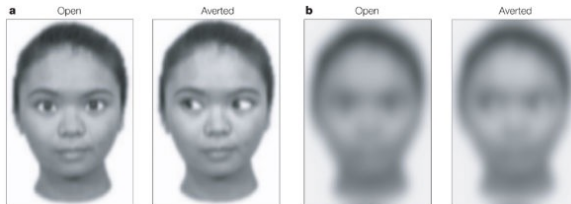
Internale Zustände müssen geschlussfolgert werden



- Potenzial ist ab der Geburt da
- Soziale Erfahrung über viele Jahre macht das Potenzial nutzbar
- Individuelle Unterschiede in Fähigkeit und Alter



Emery, 2000



Links: Sehstärke nach der Geburt

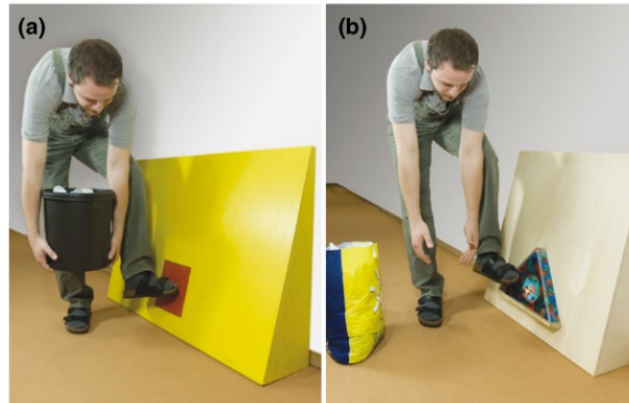
Unten: Gaze following bei 4 Monate altem Kind





Meltzoff & Moore, 1977

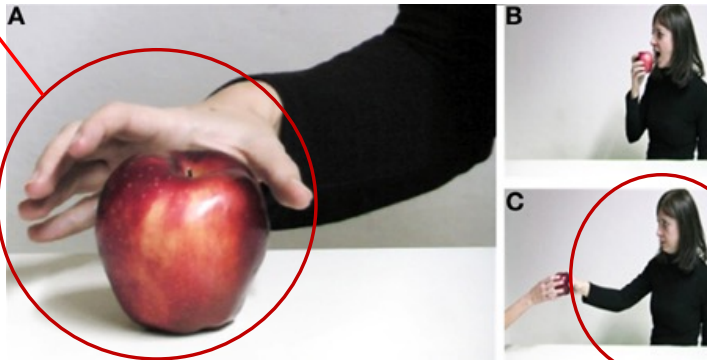
Motor Imitation



Buttelmann et al., 2007

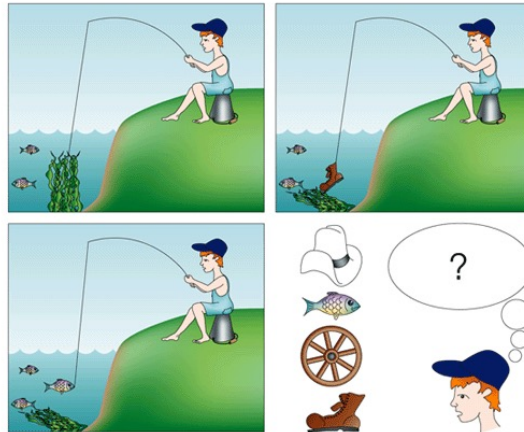
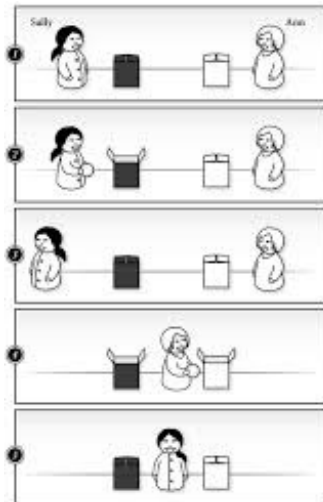
Rational Imitation

Action: Wo und was?



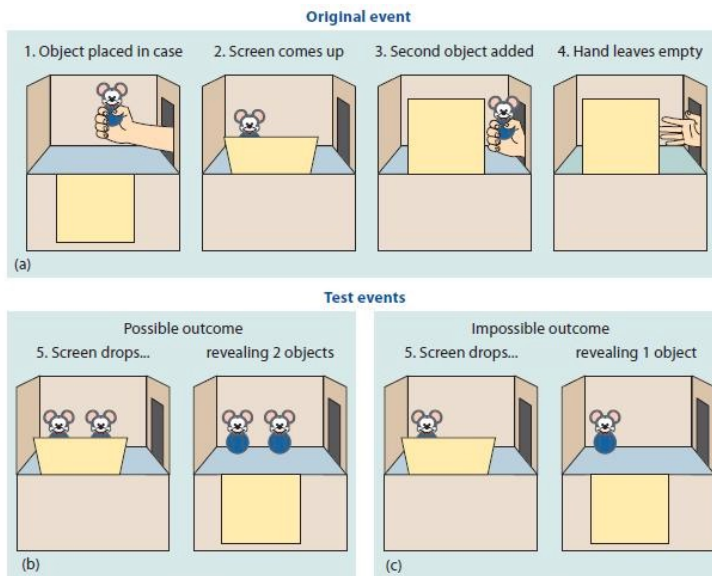
Actor: Wer und Warum?

False-Belief Aufgaben



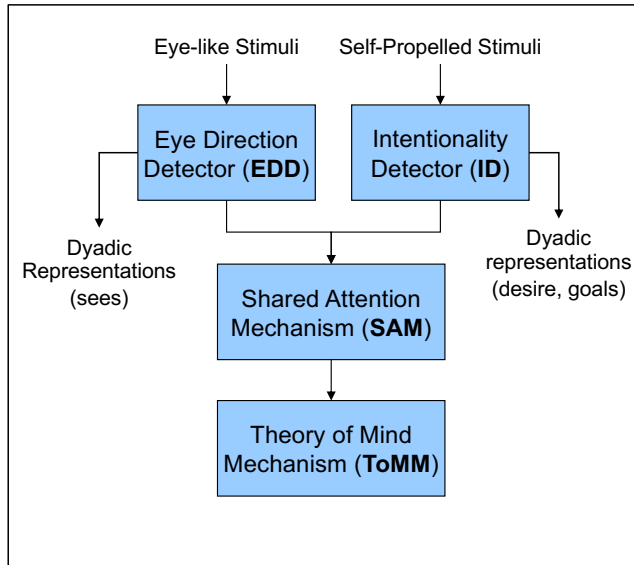
Wimmer & Perner, 1983

- Präferenz für menschliches Verhalten
- Imitationsverhalten
- Nachahmung von zielgerichteten Handlungen
- Anzahl an “pretend play”
- Verletzung von Erwartungen



Beobachtung

Längere Fixation
für unmögliche
Ergebnisse



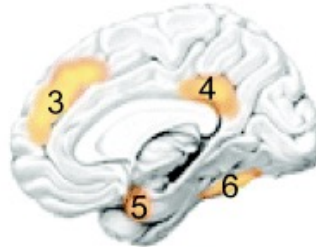
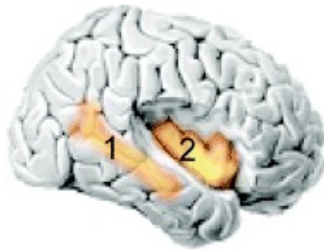
Zwei Arten von **Input**:

- Augenähnliche Stimuli (**EDD**)
- Selbstgesteuerte Stimuli (**ID**)

Zwei **kognitive Prozesse**:

- **SAM** (Aufmerksamkeit)
- **ToMM** (Schlussfolgern)

Baron-Cohen, 1985

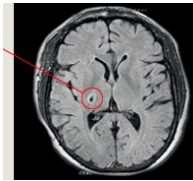


- 1 STS
- 2 Insula
- 3 mPFC
- 4 Cingulate (ACC)
- 5 Amygdala
- 6 Fusiform Gyrus
- 7 Parietal Cortex
- 8 Premotor Area
- 9 TPJ

- Kognitive Störung, bei der eine Person nicht in der Lage ist, mentale Zustände sich selbst oder anderen zuzuordnen
- Infolge dieser Störung sind sich Betroffene der mentalen Zustände anderer nicht bewusst
- Betroffene sind auch nicht in der Lage, anderen Überzeugungen und Wünsche zuzuordnen

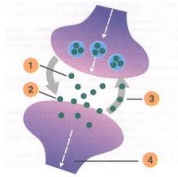
Störungsbilder:

- Autismus
- Schizophrenie
- ADHS
- Persönlichkeitsstörungen
- Demenz
- Schlaganfälle



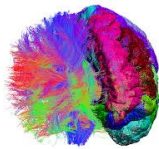
Läsionen

mPFC, STS, ACC



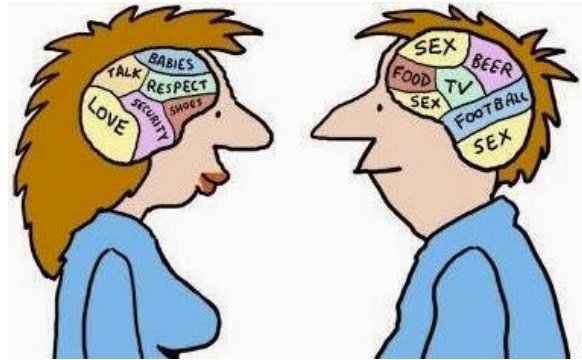
Neurotransmitterungleichgewicht

Dopamin



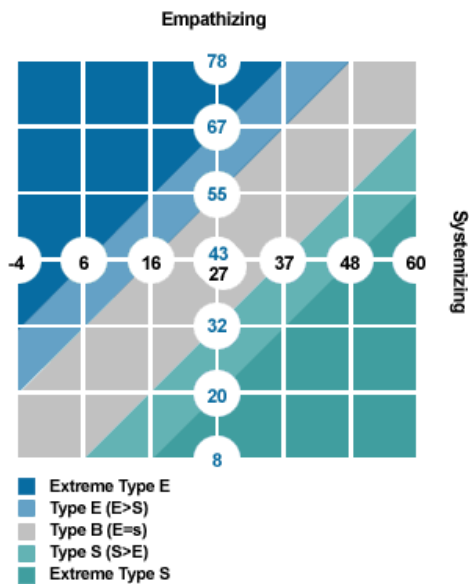
Schwache Vernetzung

Autismus Spektrum Disorder



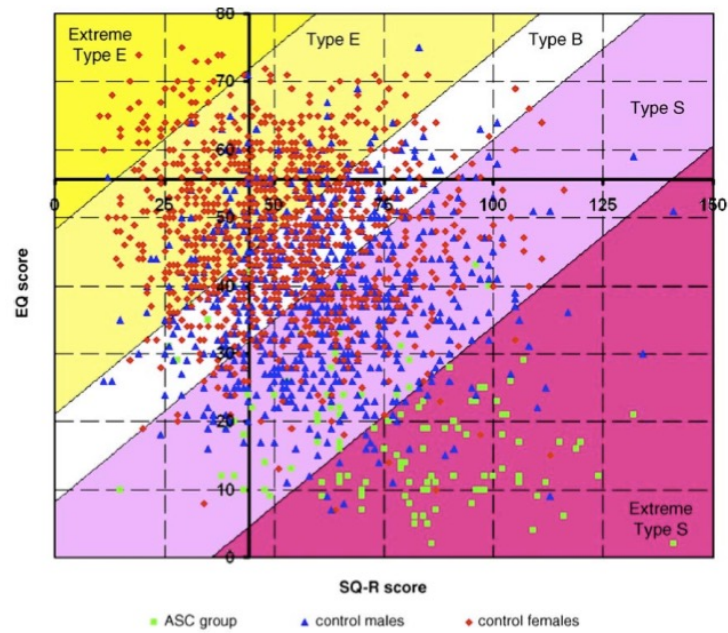
“The male brain is programmed to **systemize** and the female brain to **empathize**”

Baron-Cohen (2003)



- **Type E**
Empathy > Systemizing
- **Type S**
Systemizing > Empathy
- **Type B**
Empathy = Systemizing
- **Extreme Type E**
Empathy >> Systemizing
- **Extreme Type S**
Systemizing >> Empathy

Baron-Cohen (2003)

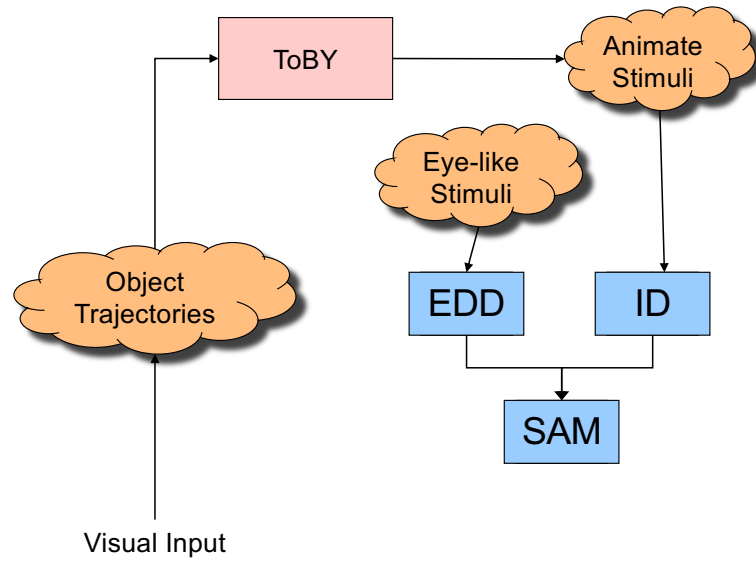


Baron-Cohen (2003)

- Theory of Mind entwickelt sich im Kindesalter, basiert auf Inferenzprozessen und ist notwendig um interne Zustände anderer zu verstehen
- Inferenzen über interne Zustände werden an Hand nonverbaler Cues (z.B. Blickrichtung, biologische Bewegung) abgeleitet
- Mind Blindness beschreibt eine Dysfunktion der Theory of Mind und ist Teil zahlreicher Störungsbilder (z.B., ASD, Schizophrenie)

Teil 3: Soziales Schlussfolgern für Roboter?



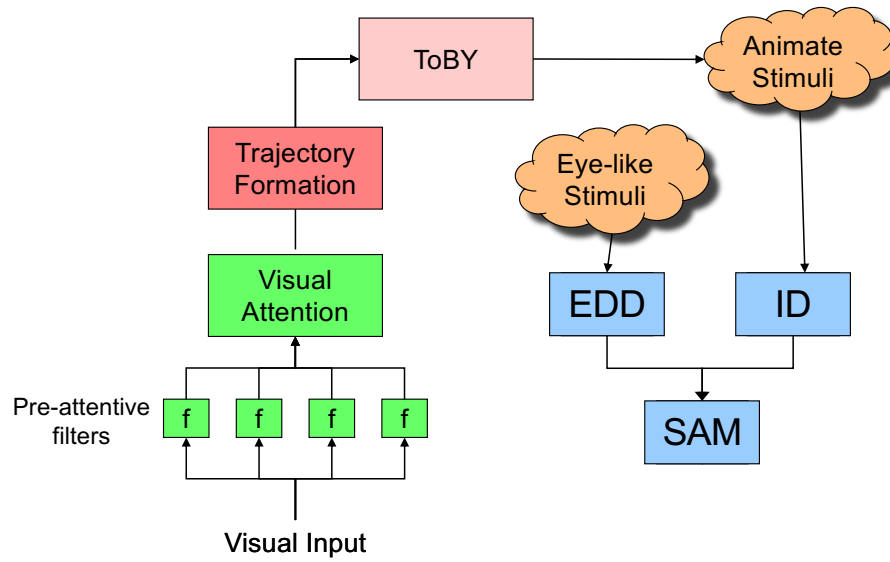


Scassellati, 2000

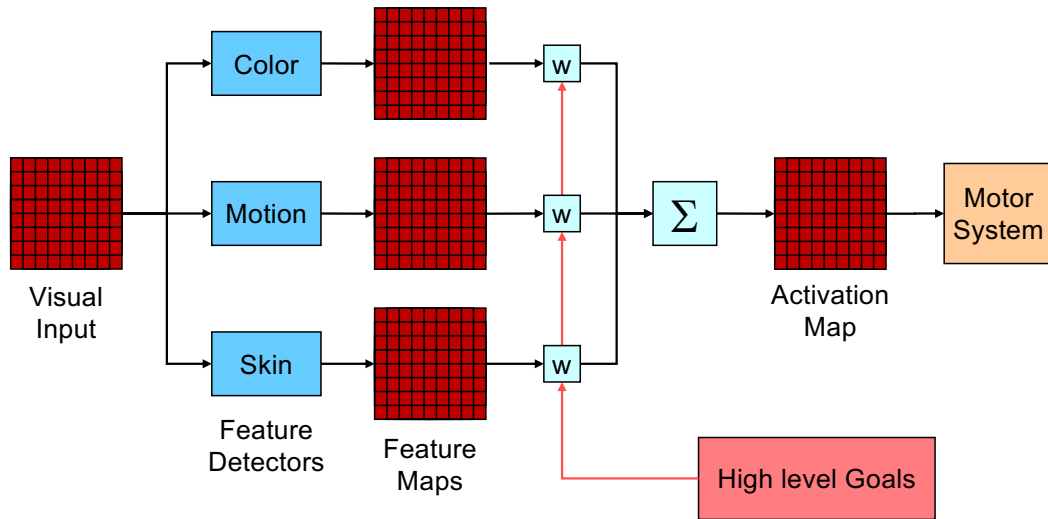




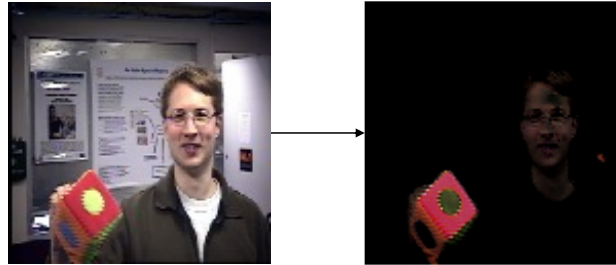
Woher weiß man, worauf die
Aufmerksamkeit gerichtet werden soll?



Scassellati, 2000



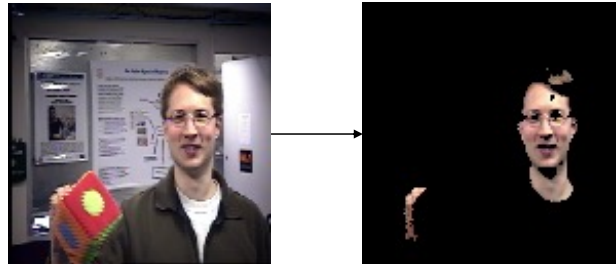
modified based on Wolfe, 1994



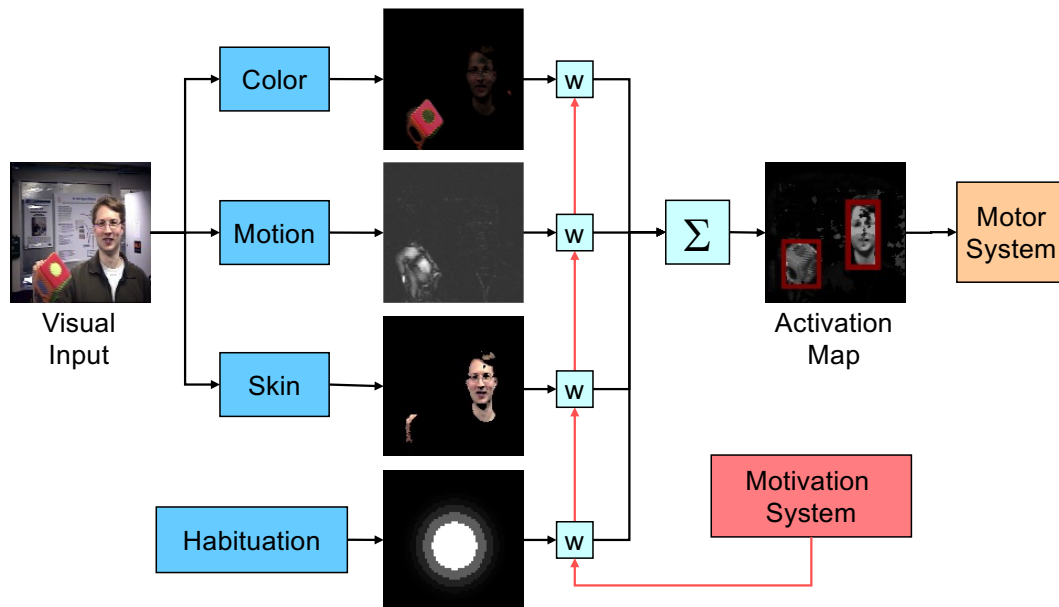
Farbe

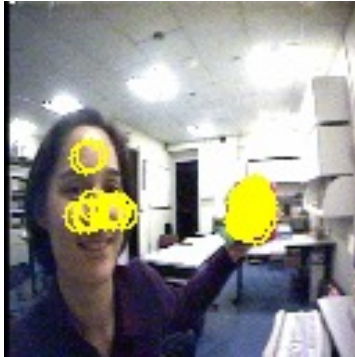


Bewegung



Haut

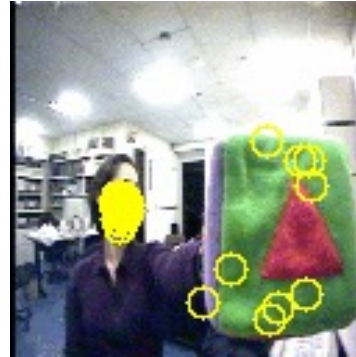




“Such das Spielzeug”

Geringe Aktivierung bei Haut, hohe Aktivierung bei gesättigten Farben

Suchzeit: 28% Gesicht, 72% Block



“Such das Gesicht”

Hohe Aktivierung bei Haut, geringe Aktivierung bei salienten Farben

Suchzeit: 28% Block, 72% Gesicht

- Internale Einflüsse beeinflussen, wie Salienz gemessen wird
- Der Roboter wird nicht durch seine Umgebung bestimmt



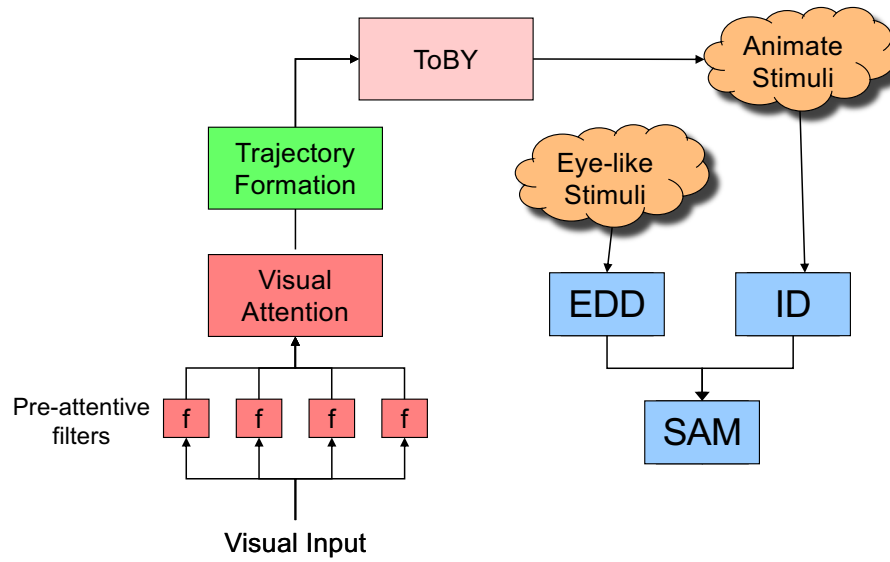
Storytelling task



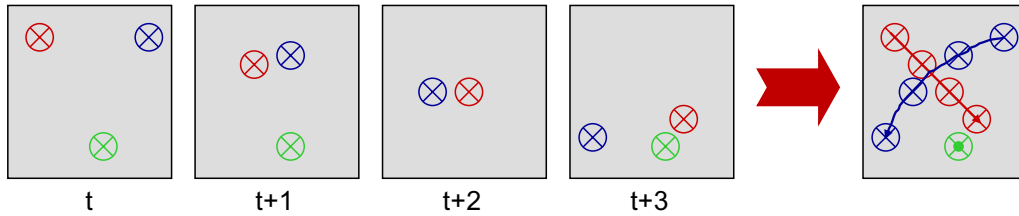
Sorting task



Aktionen verstehen

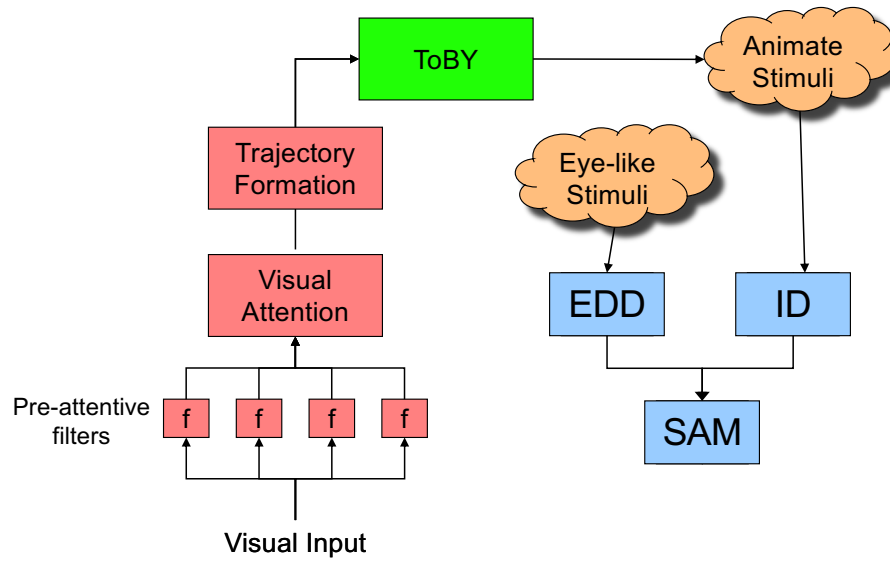


Scassellati, 2000

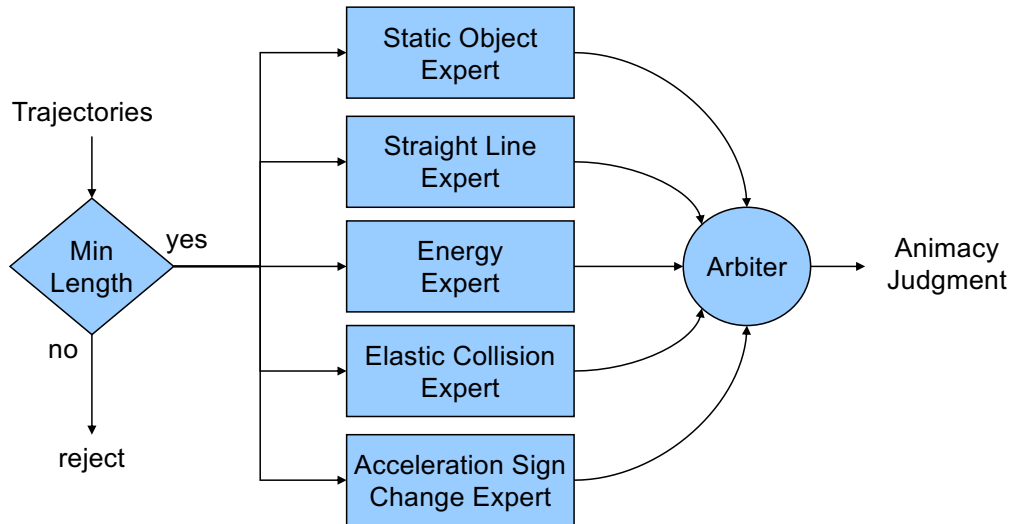


Objective is to identify sequences through a subset of the frames

$$\{P_t^{\alpha_{11}}, P_{t+1}^{\alpha_{12}} \dots P_{t+k}^{\alpha_{1k}}\}, \{P_{t+1}^{\alpha_{21}}, P_{t+1}^{\alpha_{22}} \dots P_{t+1}^{\alpha_{2k}}\}, \dots \{P_{t+k}^{\alpha_{k1}}, P_{t+k}^{\alpha_{k2}} \dots P_{t+k}^{\alpha_{kk}}\}$$



Scassellati, 2000





Lebendig

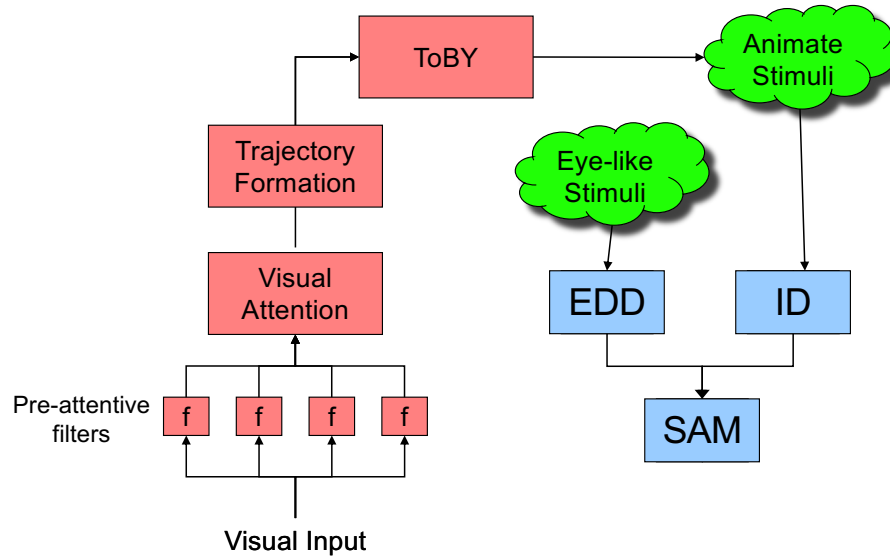


Nicht lebendig

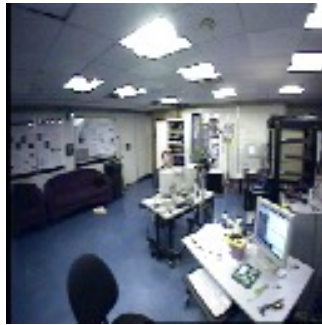


Lebendig

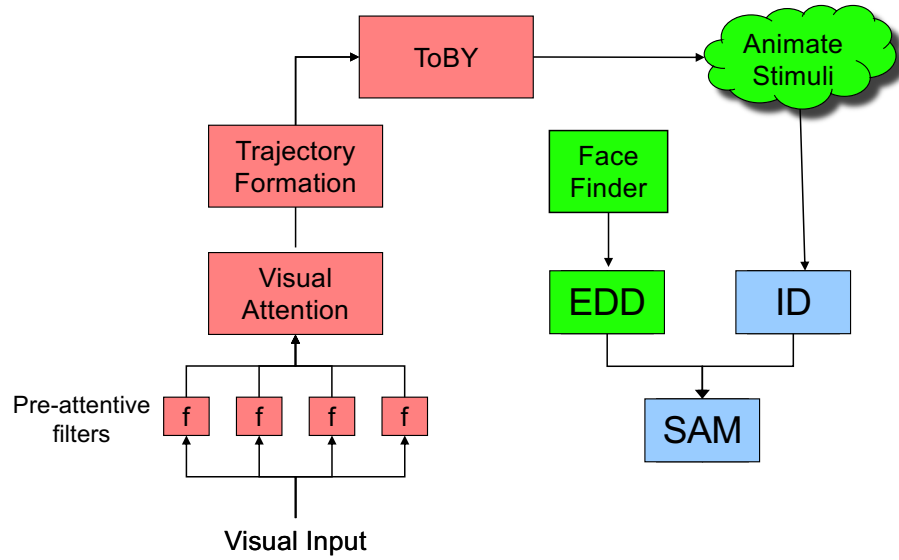




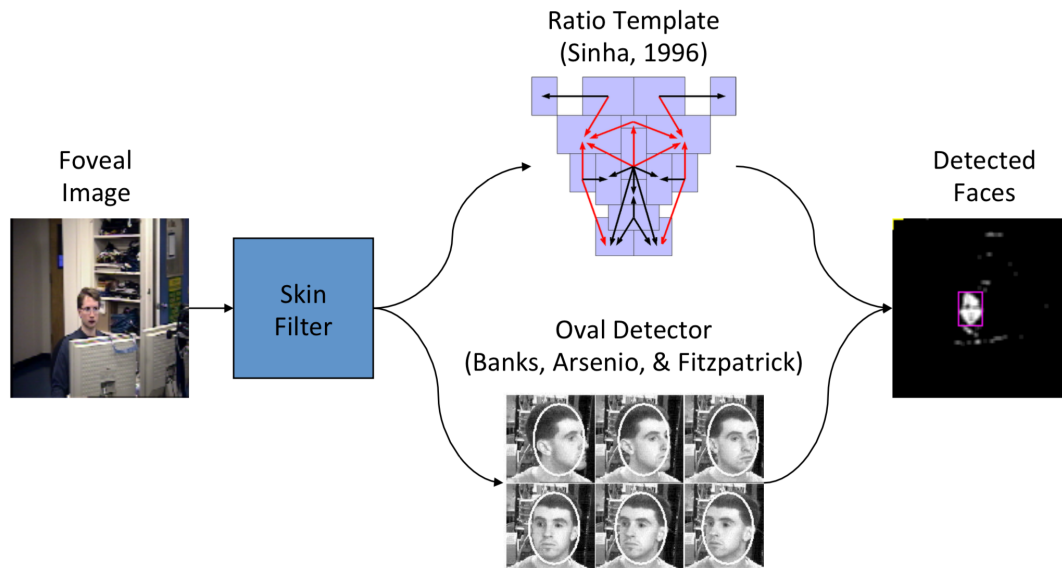
Scassellati, 2000

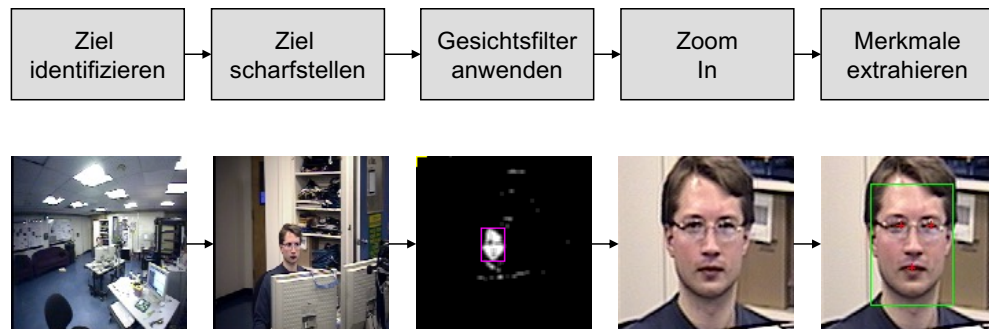


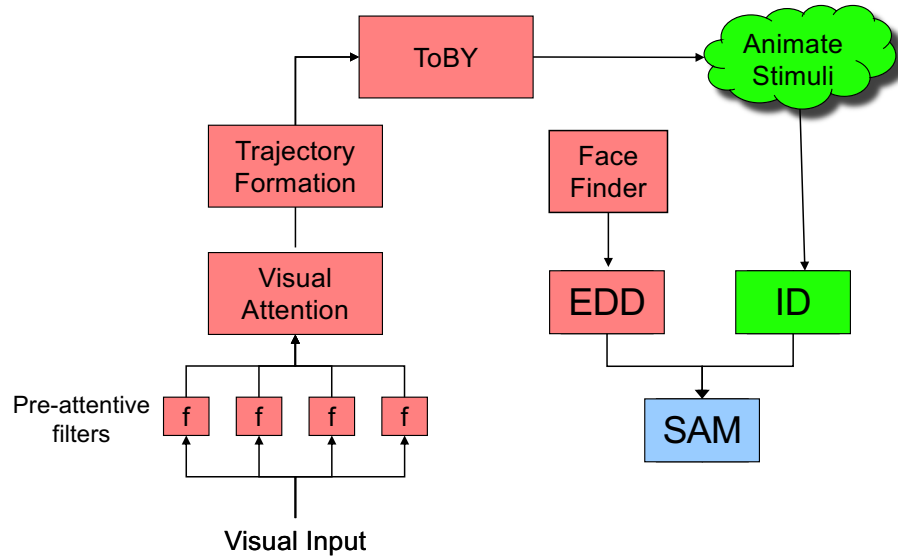
Kannst du das Gesicht im Bild finden?
Kannst du erkennen, wohin das Gesicht sieht?



Scassellati, 2000







Scassellati, 2000



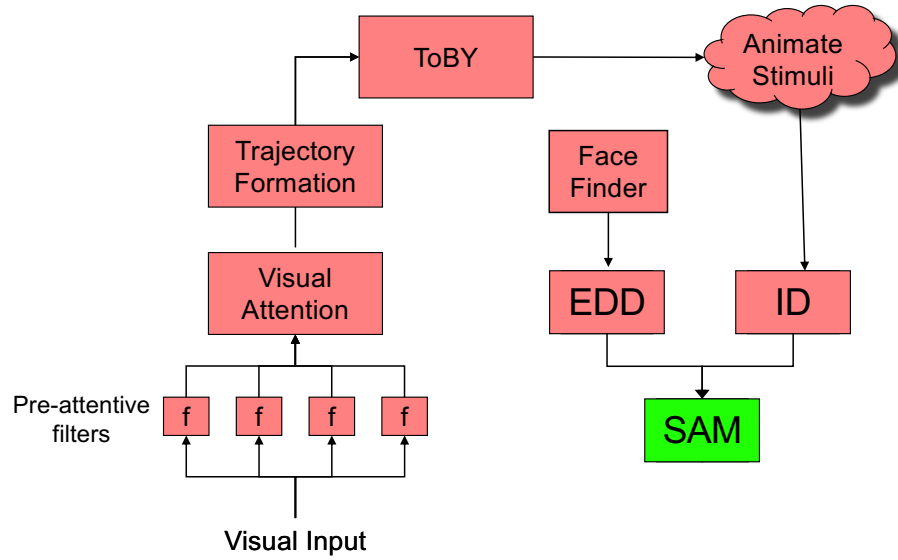
Annäherung

- Nicht-zunehmender Abstand
- Angepasster relativer Kurs

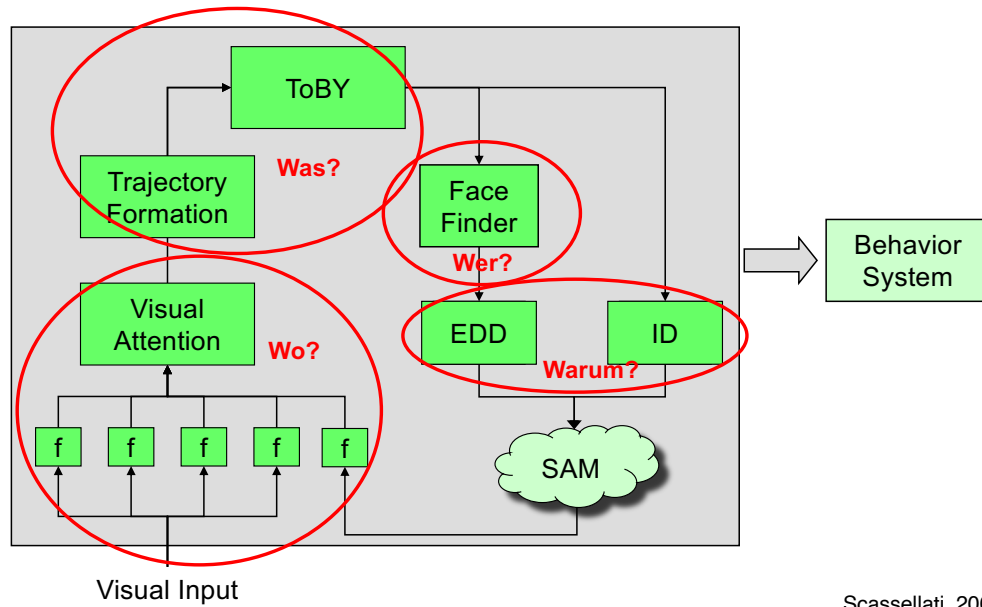


Vermeidung

- Nicht-verringender Abstand
- Gegenläufiger relativer Kurs



Scassellati, 2000



Scassellati, 2000

- Soziales Schlussfolgern und Problemlösen findet dann statt, wenn man mit Agenten konfrontiert ist, **die komplexe innere Zustände erleben** und **in anderen erzeugen können**
- **Mind perception** beschreibt das Phänomen, Andere als Agenten mit komplexen inneren Zuständen wahrzunehmen, egal ob das tatsächlich zutrifft oder nicht → Attribution bzw. Zuschreibung
- **Theory of Mind** oder mentalizing beschreibt das Schlussfolgern über die inneren Zustände Anderer. Dies beruht auf **Inferenz**
- Theory of Mind Module, die psychologisch plausibel sind, **existieren auch für Roboter**